

Convective turbulent mixing drives rapid upwelling along the ocean's bottom boundary

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Deep-ocean upwelling, driven by small-scale turbulent mixing^{1,2}, plays a central role in Earth's climate by regulating the ocean's capacity to sequester heat and carbon for decades to millennia^{3,4}. Recent theoretical studies⁵⁻⁸ have hypothesized that this upwelling may primarily occur within a thin bottom boundary layer

(BBL) adjacent to the sloping seafloor. However, little is known about the mixing processes that might drive such along-bottom upwelling, and the problem remains the subject of an enduring and vigorous debate⁸⁻¹¹. Here, we elucidate the processes governing BBL-focussed upwelling in a typical continental-slope canyon, in which very rapid upwelling is observed¹². We show that upwelling along the canyon stems from episodic cells of convective turbulent mixing up to 250 m in height, generated by tidal currents sweeping up- and down-canyon. Once per tidal cycle, these currents develop a vertical shear that conveys dense waters over slower-flowing lighter waters below, causing the dense waters to convectively mix, lighten and upwell. We argue that this upwelling mechanism is likely to be of wider representativeness, substantiating the view that deep-ocean upwelling can predominantly occur along the ocean's sloping boundaries.

Small-scale turbulent mixing has long been recognized as the key driver of deep-ocean diapycnal upwelling, through which dense abyssal waters of high-latitude origin are transformed into lighter, mid-depth waters^{1,2,5}. Starting with Walter Munk's work in the 1960s¹, it has been extensively argued that upwelling is broadly distributed across the global ocean below ~1,000 m [ref. 13-15], and that its underpinning mixing is induced principally by the breaking of internal waves radiating from rough topography^{13,16,17}. A flaw in this argument is raised, however, by observations of the deep-ocean internal wave field and associated turbulence: the intensity of such turbulence above rough topography – where the turbulence is commonly most vigorous – characteristically increases with depth^{18,19}. This implies that water parcels subject to wave breaking mix more rapidly with underlying denser layers than with overlying lighter waters, and so must become denser and *downwell* – not upwell – across the ocean's density stratification^{5,6}.

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61 A potential solution to this conundrum is provided by a body of recent theoretical
62 work⁵⁻⁸, founded on the principle that mixing must vanish at the ocean's solid
63 boundaries. Since water parcels close to the seabed may only mix with overlying lighter
64 waters, they must become lighter and *upwell* across stratification. If topography is
65 sloped, as is often the case in the ocean, such lightening of near-boundary waters could
66 result in an upward along-boundary flow that would transport those waters to shallower
67 depths. Thus, in this alternative view, deep-ocean diapycnal upwelling would be
68 expected to occur via focussed and rapid along-boundary flows within an $O(10\text{ m})$ -
69 thick, weakly stratified layer – the BBL – directed up sloping topography, with the
70 downward reduction in mixing across the BBL governing the upwelling's intensity.
71 Direct observational evidence in support of this scenario is, however, lacking, and the
72 fundamental physics of BBL mixing and upwelling remain the focus of a vibrant
73 discussion⁸⁻¹¹.

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75 To address this evidence gap, we performed a suite of measurements of turbulent
76 mixing, diapycnal upwelling and their driving processes along a paradigmatic ocean
77 boundary: a continental-slope canyon, the likes of which occupy over one fifth of all
78 continental slopes bordering the global ocean^{20,21}. The canyon targeted here is
79 approximately 30 km long and 8 km wide at its entry, and is located in the Rockall
80 Trough (Northeast Atlantic; Fig. 1a) – a basin characterized by generally moderate
81 deep-ocean stratification and circulation²² and barotropic tidal currents²³. The
82 observations were obtained in July-October 2021 under the auspices of the U.K.-U.S.
83 Bottom Boundary Layer Turbulence and Abyssal Recipes (BLT-Recipes) programme,
84 using both vessel-deployed and moored instrumentation. The ship-based measurements
85 included (Fig. 1a): an along-canyon section of 11 full-depth hydrographic and

(temperature and shear) microstructure profiles, which captures the general increase in the intensity of turbulence with proximity to the seabed (Fig. 1b) that is typical of regions of rough topography^{18,19}; 6 finely-resolved time series of hydrographic and temperature microstructure profiles within the canyon spanning at least one semidiurnal tidal cycle; and measurements of the dispersion of a fluorescein dye injected near the seabed, which revealed the occurrence of exceptionally rapid diapycnal upwelling of $O(10^{-3} \text{ m s}^{-1})$ [$O(100 \text{ m d}^{-1})$, or roughly 10,000 times the global-mean upwelling rate^{1,2,14,15}] along the canyon's thalweg¹². This diapycnal velocity is equivalent to a warming of $\sim 0.3^\circ\text{C d}^{-1}$ as water parcels upwell across the canyon's stable background stratification, which is primarily determined by temperature (Fig. 1b; see Methods). High (vertical and temporal) -resolution records of turbulent fluctuations in three-dimensional velocity and temperature were acquired by a mooring instrumented with an acoustic Doppler current profiler (ADCP), 7 modular acoustic velocity meters and 83 thermistors, distributed across the deepest 300 m of the water column. The mooring was deployed near the canyon's head (Fig. 1a) for 89 days (i.e. approximately 172 semidiurnal and 106 inertial periods), and thereby afforded robust statistical sampling of variability in turbulent parameters within the canyon across a broad range of frequencies (including tidal and near-inertial). Further details of the data set are given in the Methods.

A window into the nature of the in-canyon turbulence is provided by the vessel-based time series of hydrographic and temperature microstructure profiles, an example of which is shown in Figures 2b,c (respectively displaying potential temperature and the rate of dissipation of temperature variance, χ – a direct metric of the intensity of turbulent mixing; see Methods) contextualised with the mooring-measured along-canyon flow (Fig. 2a). The canyon's background hydrographic structure entails a

reduction in temperature with depth of approximately 1°C over 400 m (Fig. 2b), which corresponds to a buoyancy frequency of $\sim 2 \times 10^{-3} \text{ s}^{-1}$ characteristic of deep-ocean stratification elsewhere^{2,5}. This background structure is prominently modulated by a semidiurnal tide, which flows up and down the canyon with peak speeds of $\sim 0.2 \text{ m s}^{-1}$ (cf. Figs. 2a,b). As the flood tidal flow displaces isotherms upward, a layer of persistently unstable stratification (within which temperature typically increases slightly with depth; Fig. 2d) develops over 2-3 hours, reaching a height of $\sim 250 \text{ m}$ at the tide's turning. Vigorous mixing ($\chi > 10^{-8} \text{ }^{\circ}\text{C}^2 \text{ s}^{-1}$, exceeding typical ocean-interior values by over one order of magnitude) occurs along the unstable layer's rising edge (Fig. 2c), a pattern suggestive of a convective mixing scenario in which isotherms fold over to produce areas of local static instability that progressively mix the fluid from above²⁴⁻²⁶. During the ebb tide, isotherms deepen and the unstable layer thins rapidly, driving lower water column re-stratification and a confinement of elevated mixing close to the seabed.

To support the predominantly convective nature of turbulent mixing in the canyon, we next describe the mechanism via which the tide may generate unstable stratification. The along-canyon tidal flow exhibits considerable vertical shear (i.e. the tide is baroclinic; Fig. 2a), with flood and ebb tidal speeds typically peaking at $> 0.15 \text{ m s}^{-1}$ about 250 m above the seabed and reducing to $< 0.05 \text{ m s}^{-1}$ within $\sim 50 \text{ m}$ off the bottom boundary. As this vertically-sheared flow advects the background temperature distribution (which comprises approximately flat isotherms; Fig. 1b) along the canyon's sloping thalweg, it conveys cold water over warm during flood tide, triggering unstable stratification and convection some 3-6 h into that tidal phase (Figs. 2a,b); and it transports warm water over cold during ebb tide, acting to restore stably-stratified conditions (Figs. 2a,b). Thus, the occurrence of convective mixing and subsequent re-

stratification can both be attributed to differential advection of the canyon's background hydrographic structure by a vertically-sheared (i.e. baroclinic) tide, with the vertical scale of the mixing and re-stratification being determined by that of the shear. Such convective interpretation of the in-canyon mixing is directly supported by two pieces of evidence. First, an idealized analytical model of the advection of the in-canyon vertical temperature profile by the measured along-canyon flow reproduces the observed tidal cycle of destruction and re-establishment of deep water-property contrasts (see Methods and Fig. S1). And second, an appraisal of the tidal flow's susceptibility to the growth of shear instabilities^{24,25} suggests that there is no systematic correspondence between periods of intense mixing and conditions propitious to the development of shear-driven turbulence (see Methods and Figs. S2 and S5); in contrast, instances of convectively unstable temperature stratification are frequent during times of vigorous mixing (Fig. S3).

The role of convective mixing in driving the rapid diapycnal upwelling observed in the canyon¹² may be quantitatively borne out by considering the mooring-based measurements, which – on account of their long extent (89 days) and high temporal resolution (~ 1 s) – capture the variability of mixing (assessed again via a temperature-derived χ ; see Methods) over many tidal cycles. The same semidiurnal sequence of isotherm shoaling and deepening, de- and re-stratification, and enhancement and suppression of χ , seen in the vessel-based observations (Fig. 2) is apparent in the mooring records (Fig. S3 and Supp. Movie), which also expose some differences between any two tidal cycles. To draw statistically robust relationships between mixing and its tidal drivers, and elucidate implications for upwelling, we construct a phase-averaged picture of the temporal evolution of the along-canyon flow (Fig. 3a) and χ (Fig. 3b) over a semidiurnal tidal cycle (see Methods). Using potential temperature (T)

as the vertical coordinate provides a ready connection between χ and the diapycnal
 (accurately approximated as diathermal, see Methods) velocity, w_T^* , which may be
 expressed as a warming rate and defined²⁷ as: $w_T^* = \frac{1}{2} \partial(\langle \chi \rangle_{z^*} \partial z^* / \partial T) / \partial z^*$ (Fig. 3c).
 Here, z^* is the distance in the diathermal direction, $\langle \chi \rangle_{z^*}$ represents an average over an
 isothermal surface in a bounded volume, and w_T^* has units of $^{\circ}\text{C s}^{-1}$. This definition
 synthesises the notion that a water parcel mixing with warmer waters more strongly
 than with colder waters will upwell, and *vice versa*^{5,6}. In our application, we assume by
 necessity that, at any phase in the tidal cycle, z^* and $\langle \chi \rangle_{z^*}$ are well approximated by the
 local vertical direction and phase-averaged χ estimated from the mooring observations,
 respectively. See Methods for a discussion of the reasonability of these approximations.

The phased-averaged view demonstrates how, in a characteristic tidal cycle, isobaths
 progressively converge in temperature space (denoting de-stratification; Fig. 3a) and χ
 is elevated across a wide range of temperatures (Fig. 3b) when the flow ~ 200 m above
 the seabed is directed up-canyon relative to the near-bottom flow, i.e. precisely when
 vertically-differential along-canyon advection is expected to induce convective
 conditions. This occurs at the tide's turning from ebb to flood (around a phase of 2 rad,
 when the flows far above and near the seabed are oriented oppositely; see Methods)
 and during flood tide (in the phase range of 2.5-5 rad, when flow is up-canyon at all
 temperatures). The causal relationship between differential advection and χ also
 transpires in the latter's diathermal distribution: χ peaks along the band of maximum
 up-canyon differential advection, some ~ 120 m above the seabed (Fig. 3b), and decays
 toward colder temperatures from there, since the intensity of up-canyon flow varies
 little over the coldest waters, i.e. in the deepest ~ 80 m. Subsequently, as the tide turns
 to ebb (in the approximate phase ranges of 0-1.5 rad and >5 rad), isobath divergence
 and a reduction in χ signal re-stratification and suppression of turbulence.

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191 The diapycnal circulation induced by the in-canyon turbulent mixing is illustrated by
192 the phase-averaged distribution of w_T^* (Fig. 3c). During flood tide, the preferential
193 elevation of χ within intermediate temperature classes (around 5°C; Fig. 3b) results in
194 diapycnal upwelling (i.e. warming) of colder waters (typically within ~120 m of the
195 seabed) at rates approaching or exceeding $10^{-5} \text{ }^\circ\text{C s}^{-1}$, and in somewhat less intense
196 diapycnal downwelling (i.e. cooling) of warmer waters aloft. Little diapycnal flow
197 occurs during ebb tide. Upon averaging over the tidal cycle to estimate the net diapycnal
198 velocity profile (Fig. 3d), waters colder than ~5°C are found to upwell at a rate of up to
199 $\sim 3 \times 10^{-6} \text{ }^\circ\text{C s}^{-1}$ (or $\sim 0.3 \text{ }^\circ\text{C d}^{-1}$), and warmer waters to downwell at a weaker rate of $\sim$
200 $1 \times 10^{-6} \text{ }^\circ\text{C s}^{-1}$ (or $\sim 0.1 \text{ }^\circ\text{C d}^{-1}$). The net upwelling rate of the colder waters
201 approximately matches the dye-derived diapycnal flow of $\sim 0.3 \text{ }^\circ\text{C d}^{-1}$ [ref. 12] – thereby
202 endorsing the role of convective mixing in driving the rapid along-boundary upwelling
203 measured within the canyon.

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205 The efficacy of convective turbulence as an agent of intense diapycnal upwelling is
206 contingent on its large vertical scale, which leads to the swift mixing of waters spanning
207 a substantial range of temperatures. In our canyon, where convection is observed to
208 reach up to ~250 m above the seabed, such vertical scale is determined by that of the
209 along-canyon flow's vertical shear – as it is this shear that generates convective
210 conditions by differentially advecting the background temperature profile (Fig. S1).
211 Our analysis (see Methods and Fig. S4) suggests that the in-canyon vertical shear is
212 associated with baroclinic tidal waves that develop within the canyon^{28,29}. The
213 characteristic vertical scale of the waves is on the order of the canyon's vertical extent,
214 which thus sets the size of the convective overturns underpinning along-slope

upwelling. The detailed physics of the baroclinic tidal waves will be described in a future study.

In conclusion, our observations of a continental-slope canyon at the edge of a North Atlantic basin provide evidence of very rapid along-boundary upwelling driven by vertically-extensive convective turbulent mixing, itself generated by semidiurnal baroclinic tidal flows sloshing up- and down-canyon (Fig. 4). The cornerstone of the upwelling mechanism is the sheared structure of the tidal waves that develop within the canyon. This shear causes relatively dense waters aloft to overshoot, and hence convectively mix with, lighter near-bottom waters during flood (Fig. 4a) – leading to diapycnal upwelling of the dense waters. Reversal of the sheared tidal flow during ebb tide induces re-stratification within the canyon via the differential down-canyon advection of near-boundary waters (Fig. 4b), and promotes the offshore export of waters that were newly mixed and upwelled along topography during the previous flood tide¹². Such ebb-time re-stratification and export reset the canyon for the next flood stage of upwelling, and communicate the effects of in-canyon mixing and upwelling toward the basin’s interior.

Although the technological complexity of our measurements inevitably confines our study to a single small domain, there is evidence to suggest that our findings may have wider representativeness along many of the ocean’s sloping boundaries. Such evidence comes in two forms. First, canyons of a similar nature to that of the one targeted here abound around the global continental slope. Our canyon’s length (~30 km), width (~8 km at its mouth), vertical extent (>1000 m from mouth to head), aspect ratio (~80°, denoting a relatively narrow canyon), V-shaped geometry (with axial slope near M2 tidal criticality, and steep, supercritical sidewalls), and characteristic barotropic tidal

speeds ($\sim 0.08 \text{ m s}^{-1}$) – all factors likely to shape the governing physics and intensity of the in-canyon mixing^{30,31} – are in line with those of many continental-slope canyons in the global ocean^{20,21,30}. Second, observational and modelling investigations of near-boundary mixing in a range of oceanic^{26,33-35}, limnic^{36,37} and idealized geophysical fluid^{38,39} settings indicate that the convective process documented here may be of broad generality to oscillatory flows over sloping topography, regardless of the details of the flow or topographic configuration. We can thus surmise that the mechanism of rapid, convectively-induced along-boundary upwelling highlighted by our work is likely to operate elsewhere. On this basis, our results provide observational support for the new paradigm of BBL-focussed deep-ocean upwelling, and point to a hitherto-overlooked process as an important agent of such upwelling.

Methods

BLT-Recipes data set. A set of targeted, vessel-based measurements of the hydrographic, velocity and (shear and temperature) microstructure properties of the ocean within and above a continental-slope canyon in the Rockall Trough (Northeast Atlantic; Fig. 1a) was acquired during expedition DY132 of the *RRS Discovery* between 19th June and 29th July 2021, supported by the U.K.-U.S. Bottom Boundary Layer Turbulence and Abyssal Recipes (BLT-Recipes) programme. See Figure 1 for a map overview of the measurements pertinent to this article.

These ship-based observations were collected with two instrument suites. First, the general along-canyon distributions of stratification and mixing were documented with a transect of 11 full-depth stations along the canyon's thalweg. At each station, vertical profiles (down to 5-50 m above the ocean floor) of hydrographic variables and velocity were recorded with: (i) a conductivity-temperature-depth / 300-kHz lower acoustic

Doppler current profiler (CTD/LADCP) package lowered from the vessel; and (ii) the untethered, free-fall Woods Hole Oceanographic Institution's High Resolution Profiler (HRP)⁴⁰ or Rockland Scientific Vertical Microstructure Profiler 6000 (VMP-6000)⁴¹, which additionally measured shear and temperature microstructure – from which the rates of dissipation of turbulent kinetic energy (ϵ) and temperature variance (χ) can be respectively quantified⁴². Second, 6 finely-resolved (with a characteristic temporal resolution of ~15 min) time series of hydrographic and temperature microstructure profiles spanning 1-2 semidiurnal tidal cycles were obtained within the canyon, over a height-above-bottom range of approximately 0-400 m, using two Scripps Institution of Oceanography-produced, lightly-tethered ultra-fast profilers equipped with a CTD and either temperature microstructure sensors (the FastCTD) or shear and temperature microstructure sensors (the Epsilometer)⁴³. The FastCTD, which was also integrated with a fluorometer, enabled monitoring of the dispersion of a fluorescein dye injected near the seabed, from which evidence of very rapid diapycnal upwelling along the canyon is drawn and discussed in a ref. 12. Full details of the dataset acquisition and processing may be found in the DY132 cruise report⁴⁴.

Vessel-based measurements were supplemented by moored observations of turbulent fluctuations in three-component velocity and temperature, obtained with a mooring instrumented with 7 modular acoustic velocity sensor (MAVS) current meters (not explicitly considered here), 83 fast-sampling thermistors and an ADCP. The mooring was deployed near the canyon's head (Fig. 1a) on 7th July 2021, during the DY132 expedition, and recovered on 4th October 2021 in cruise DY138 of the *RRS Discovery*. The fast-sampling thermistors were placed at intervals of 1-5 m (with higher vertical resolution near the seabed) over the height-above-bottom range of 2-300 m, and were programmed to sample at 1 Hz over a period of 89 days. A downward-looking 75 kHz

RDI ADCP was deployed in a float at the top of the mooring, at a height above bottom of 300 m, and sampled with a vertical bin size of 16 m every 15 minutes. Thus, the moored measurements were performed at a substantially finer temporal resolution, and during a much longer period – extending over ~172 semidiurnal tidal cycles and ~106 inertial cycles – than the ship-based observations. As such, the moored sensors sampled the variability in turbulent parameters, and its relationships with the evolving background stratification and flow, with far greater statistical robustness than provided by the vessel-based data.

Calculation of turbulent kinetic energy dissipation rates from shear microstructure measurements. The rate of dissipation of turbulent kinetic energy, ε , was computed from shear microstructure measurements obtained with three different profilers (HRP, VMP-6000 and Epsilon; see Methods), which recorded vertical gradients in horizontal velocity on centimetre scales at 512 Hz to within typically 2 m of the ocean floor. Data processing was conducted using algorithms developed originally for the HRP^{40,41,43,44}. The variance in the shear in the vertical direction, $\overline{(\partial u / \partial z)^2}$, was calculated every 0.5 m, using shear spectra computed over bin widths of 1 s and integrated between 1 Hz and the spectral minimum in the 10–25 Hz band, or the 25–100 Hz band for $\varepsilon > 10^{-7} \text{ W kg}^{-1}$. Values of ε were calculated from the variance of the shear as $\varepsilon = 7.5 \nu \overline{(\partial u / \partial z)^2}$, where ν is the molecular viscosity of seawater and isotropy is assumed⁴².

Calculation of turbulent mixing rates from temperature microstructure measurements. The rate of dissipation of temperature variance, χ , was computed from temperature microstructure measurements acquired with four different profilers (HRP, VMP-6000, FastCTD and Epsilon; see Methods). Similarly to ε , data processing

was conducted using algorithms developed originally for the HRP^{40,41,43,44}. The variance of the vertical temperature gradient, $\overline{(\partial T / \partial z)^2}$, was calculated every 0.5 m, using bin widths of 1 s. Each profiler carried two fast-response thermistors, and unless one sensor was deemed noisy, the mean of the χ estimates from each sensor pair was used. Values of χ were computed from the variance of the vertical temperature gradient as $\chi = 6\kappa\overline{(\partial T / \partial z)^2}$, where κ the molecular diffusivity of seawater and isotropy is assumed⁴². Prior to the computation of χ , $\overline{(\partial T / \partial z)^2}$ was corrected for nonlinearity and pressure dependence in the thermistors' response by reference to the temperature measurements obtained with a CTD mounted on each profiler. Microstructure cast locations were recorded as the midpoint between the instrument deployment and recovery positions in each station (HRP and VMP-6000), or as the vessel's position at the time of sampling (FastCTD and EpsiFish).

Fundamentally, χ represents the rate of dissipation of temperature variance by molecular mixing. Since this temperature variance is produced by centimetre-scale turbulent diathermal motions acting on the evolving temperature field, χ provides a metric of the diathermal mixing effected by such motions²⁷. In the context of our canyon data set, the diathermal and diapycnal directions are closely aligned, as evidenced by the tightness of the in-canyon potential temperature-density relationship (Fig. 1b). Thus, χ adequately quantifies the intensity of turbulent diapycnal mixing within the canyon.

Calculation of turbulent mixing rates from mooring-based temperature observations. The rate of dissipation of temperature variance, χ , was computed from temperature time series acquired with each of the 83 fast-sampling (1 Hz) thermistors

in the mooring. χ can be estimated from temperature records by fitting a theoretical model of inertial subrange turbulence to measured temperature spectra⁴⁵⁻⁴⁷,

$$\phi_T(k) = C_\theta \chi \varepsilon^{-\frac{1}{3}} k^{-\frac{5}{3}},$$

where $\phi_T(k)$ is the horizontal wavenumber spectrum of temperature, k is the horizontal wavenumber, and C_θ is the Obukhov-Corrsin constant (taken as 0.4). This wavenumber inertial subrange spectral model is translated to frequency, σ , via Taylor's frozen field hypothesis, $\sigma = Uk$, with U as the magnitude of the flow past the sensor^{46,47}. Relating the rate of dissipation of turbulent kinetic energy, ε , to a buoyancy flux⁴⁶, and assuming the buoyancy frequency to be a function of temperature only, then yields:

$$\chi = \left[\left(\frac{g\alpha}{2\Gamma \frac{\partial T}{\partial z^*}} \right)^{\frac{1}{3}} \left(\frac{2\pi}{U} \right)^{\frac{2}{3}} \sigma^{-\frac{5}{3}} \frac{\phi_T(\sigma)}{C_\theta} \right]^{\frac{3}{2}},$$

where g is gravity, α is the thermal expansion coefficient, and Γ is the dissipation ratio. This expression enables χ to be calculated from theoretical fits to the observed temperature spectra. In our implementation, we computed spectra in 5-minute segments, scaled them by $\sigma^{5/3}$, and averaged the (fitted and scaled) spectra for periods of 10-100 s (or 10- $[hab/U]$ s for instruments whose height above bottom (hab) was sufficiently small that $hab/U < 100$ s) representative of the inertial subrange. This produced an estimate of χ every 5 minutes for each thermistor. Γ was taken as 0.2 (a common value in much of the ocean)⁴⁸, but the sensitivity of χ to this choice was explored by repeating the calculation with $\Gamma = 0.4$ (a value that is likely more appropriate to our canyon⁴⁹).

A number of practical computation choices must be made in estimating χ from thermistor measurements, including: the window size and shape implemented in the Welch spectral calculation; the frequency range used to average the temperature

spectra; the application of the mean or median operators in averaging over this frequency range; and the aforementioned selection of a Γ value. We performed a broad suite of sensitivity tests using a representative sample of the entire thermistor data set, in order to assess the robustness of our χ estimates. The resulting χ data were only weakly sensitive to reasonable variations in computation choices. Thus, changes in spectral window size and shape, and in mean vs. median operator choice, produced a change in χ of at most a factor of 1.5. Similarly, χ also exhibited weak sensitivity to the choice of frequency range: χ values changed by less than a factor of 1.5, except when excessive narrowness of the frequency range increased the impact of spectral noise. Varying Γ from 0.2 to 0.4 resulted in a reduction in χ by a factor of 1.4. Finally, we conducted a thorough examination of 5-min-averaged temperature spectra for each thermistor to ensure that the selected frequency range is representative of the inertial subrange, with spectra consistently exhibiting the typical $-5/3$ slope.

The estimates of χ obtained from the moored thermistors are of a subtly different nature to those acquired by the microstructure profilers – a distinction that has both advantages and challenges for our work. Profiler-based χ measurements are obtained at the closing stages of the turbulent cascade of temperature variance, i.e. near the dissipative (Kolmogorov) scale at which mixing takes place. Thus, they offer an accurate, relatively assumption-free view of local mixing. In contrast, mooring-based χ values are derived from observations within the inertial subrange, such that the measured temperature variance is yet to cascade further downscale before reaching the dissipative scale. A simple estimate of the time scale of the downscale cascade ($\langle T'^2 \rangle / \chi$, where $\langle T'^2 \rangle$ is the characteristic thermistor-measured temperature variance on inertial-subrange time scales) suggests that it may take up to several hours for that variance to be dissipated. As a result, mooring-based χ values provide an assumption-heavier, but

spatially-integrated metric of the in-canyon mixing. The integral character of mooring-based χ estimates is arguably advantageous to our endeavour of quantifying upwelling from mixing rates, as it reduces the sensitivity of the calculation to the precise location of the mooring – which effectively samples mixing conditions along a substantial segment of the canyon, spanning an along-canyon advective length scale corresponding to $\langle T'^2 \rangle / \chi$ of O(100-1000 m).

A further advantage of mooring measurements is that they are approximately Eulerian, and thereby provide unbiased sampling of all phases of the tidal cycle. Whereas the mooring's uppermost instruments move horizontally by only <50 m during one such cycle, microstructure profilers may drift with the tidal currents by as much as ~3 km in the course of a cast⁴⁹. This drift is expected to curtail the extent to which profilers may sample the most vigorous mixing in a tidal cycle, which takes place immediately below the strongest up-canyon flows (cf. Figs. 3a-3b). As a profiler descends into these intense flows, it will tend to be advected away from the highly-turbulent layers below, potentially introducing a detrimental sampling bias.

Assessment of the relative roles of convective and shear-driven turbulence in driving mixing. Turbulent diapycnal mixing in the stratified ocean interior is most commonly effected by shear-driven turbulence⁵⁰⁻⁵². Such turbulence ensues when a stably stratified fluid with buoyancy frequency $N = \sqrt{-(g/\rho)(\partial\rho/\partial z)}$ (with ρ as density) and subjected to a steady flow with vertical shear $S = \partial U / \partial z$ (with U as flow speed) becomes unstable to shear instability, which grows as the gradient Richardson number $Ri = N^2 / S^2 < 0.25$ (refs. 24, 25, 53, 54). This criterion may be relaxed somewhat to $Ri \sim 1$ under a variety of conditions, e.g., when the fluid is viscous-diffusive⁵⁵, the vertical shear is unsteady⁵⁶, or there is pre-existing turbulence in the

fluid⁵⁷. The extent to which shear-driven turbulence can explain our observations of a semidiurnal cycle in turbulent mixing may be assessed by computing the gradient Richardson number from the moored measurements of temperature and horizontal velocity. Specifically, N^2 is calculated from thermistor data as $N^2 = g\alpha \frac{\partial T}{\partial z}$; and S^2 is quantified from the ADCP record of horizontal velocity as $S^2 = (\partial u / \partial z)^2 + (\partial v / \partial z)^2$, where (u, v) is the horizontal velocity vector. These computations are performed in vertical bins of 16 m and in temporal bins of 15 min (both in line with ADCP bin sizes). Their results (Figs. S2 and S5) reveal that there is no systematic coincidence between intense turbulent mixing events and near-critical or reduced Ri values approaching $Ri \sim 1$, suggesting that shear instabilities are not the primary driver of the observed turbulence.

Our recurrent observation of vigorous mixing ($\chi > 10^{-8} \text{ }^\circ\text{C}^2 \text{ s}^{-1}$) along the rising edge of a layer with persistently unstable stratification during flood tide (Figs. 2 and S3) indicates that such mixing may be underpinned by convective – rather than shear-driven – turbulence^{25,26}. Convective turbulence in the interior of a stratified water column stems from a local instability in the vertical density gradient generated by the advection of relatively dense water over lighter water, and results in the progressive mixing of the unstable patch from above^{25,26,33-36}. Although such a convective arrangement may arise ephemerally within individual shear instability events (i.e. during shear-driven turbulence)^{38,39, 58}, here we use the term ‘convective turbulence’ more restrictively, to refer to the turbulence occurring in association with unstable stratification generated by a large-scale and persistent sheared flow with vertical and temporal scales (~ 250 m and several hours, respectively; see e.g., Fig. 2a) greatly exceeding those characteristic of the shear instability process [$O(1-10 \text{ m})$ and $O(1-10 \text{ min})$ for typical ocean-interior conditions]^{24,58}.

446

447 In order to quantitatively assess whether the measured mixing is convective, we
448 construct a simple two-dimensional (along-canyon distance versus depth) analytical
449 model in which the in-canyon potential temperature distribution, which characterises
450 the in-canyon stratification, is advected by the observed along-canyon flow over a
451 characteristic semidiurnal tidal cycle. The in-canyon temperature field is assumed to be
452 a sole function of depth, $T(x, z) = T(z)$ (with x as the along-canyon distance), i.e. mean
453 isotherms are taken as approximately flat within the canyon in line with our
454 measurements (Fig. 1b), and is initialised from mooring observations at the time of ebb-
455 to-flood tide reversal. In turn, the horizontal component of the along-canyon flow u is
456 approximated as a function of height above the local bottom (hab) and time (t), i.e.
457 $u(x, z, t) = u(hab(x), t)$, with the flow's vertical component assumed to be vertically
458 uniform within the canyon and parallel to the canyon's thalweg, i.e. $w(x, t) =$
459 $\eta u(x, z, t)$, where $\eta \approx 0.07$ is the bottom slope of the canyon's thalweg. This set of
460 approximations assumes that (i) the along-canyon velocity profile measured by the
461 mooring is widely representative along the canyon's thalweg, and (ii) the component
462 of the flow normal to the seafloor plays a modest role in the advection of temperature
463 within the canyon. Some support for (i) is provided by the recurrent pattern of
464 isothermal displacement and χ over a semidiurnal tidal cycle seen at a range of locations
465 along the canyon (e.g., compare Figures 2 and S2, constructed from measurements
466 acquired 5 km apart). Regarding (ii), the moored observations of along-canyon and
467 vertical velocity (Fig. S5) indicate that particle trajectories are up to 40% steeper than
468 the canyon's thalweg during flood tide, such that our model represents a conservative
469 assessment of the potential for convective conditions to occur. Assumption (ii)
470 effectively implies the neglect of the potential role of nonlinear internal wave
471 deformation and breaking in triggering convective mixing³⁴.

472

473 The model is integrated in time over a semidiurnal (flood followed by ebb) period and
474 an effectively unbounded along-canyon domain, in order to estimate the depth from
475 which waters sampled by the mooring at each time originated (Fig. S1). A model
476 integration for the same period as shown in Figure 2a is considered here, but other
477 periods produce very similar results. During the flood phase of the tide, water parcels
478 translate a characteristic up-canyon distance of ~ 2.5 km at ~ 250 m above the seafloor
479 but only < 1 km within ~ 80 m of the seafloor, i.e. waters aloft move up-canyon by > 1.5
480 km relative to near-bottom waters (not shown). With along-canyon parcel trajectories
481 assumed parallel to the seafloor, the differential up-canyon motion of waters with a
482 substantial height above bottom relative to those below causes waters arriving at the
483 mooring's deepest ~ 250 m during flood tide to originate from a narrow range of depth
484 (and therefore temperature) horizons (Fig. S1). Such homogenization of deep water
485 properties generates the conditions necessary for convective turbulence to ensue^{26,35,36},
486 suggesting that the mixing observed in the canyon may be primarily of a convective
487 nature. This suggestion is reaffirmed by the favourable comparison of the model's
488 predictions with the observed temperature record (Fig. S1), which indicates that the
489 measured emergence of a thick, homogeneous BBL during flood tide stems from all
490 BBL waters having been sourced over a narrow range of depths upstream. The converse
491 scenario is found to occur during ebb tide, for which BBL thinning coincides with a
492 progressive sharpening of the vertical gradient in the source depth of near-bottom
493 waters.

494

495 Further endorsement of a convective interpretation of the in-canyon mixing may be
496 obtained on energetics grounds. In order for convective turbulence to develop,
497 differential advection must generate convective available potential energy at a rate

matching that of turbulent kinetic energy production via a buoyancy flux^{26,33,35}, given by $J_b = \frac{g}{\rho} \overline{w'\rho'} \approx -\frac{g}{\rho} h \Delta u \frac{\partial \rho}{\partial x}$, where h is the height of the convectively driven layer, Δu is the along-canyon velocity difference across that layer, and $\frac{\partial \rho}{\partial x}$ is the up-canyon density gradient parallel to the seafloor within the layer. Taking values typical of our measurements ($h \sim 100$ m, $\Delta u \sim 0.1$ m s⁻¹, and $\frac{\partial \rho}{\partial x} \sim -10^{-5}$ kg m⁻⁴; Figs. S2 and S3) yields a rate of generation of convective available potential energy of $\sim 10^{-6}$ m² s⁻³. This is on the same order as the buoyancy flux estimated from our microstructure and hydrographic observations, given by $\frac{\chi}{2\left(\frac{\partial T}{\partial z}\right)^2} N^2 \sim 5 \times 10^{-7}$ m² s⁻³ (ref. 26) using $\chi \sim 2 \times 10^{-6}$ °C² s⁻¹ (Fig. 3b), $\frac{\partial T}{\partial z} \sim 2 \times 10^{-3}$ °C m⁻¹ (Fig. 1b) and $N^2 \sim 2 \times 10^{-6}$ s⁻² (Fig. S5). Such broad correspondence supports the plausibility of the observed turbulence being convectively driven by the along-canyon flow.

Finally, additional indication of the convective nature of near-boundary turbulence within the canyon is provided by frequency spectra of temperature variability measured by a moored thermistor string located ~ 1.5 km down canyon from that considered in this work⁵⁹. Analysis of those data reveals that, in the deepest ~ 30 m of the canyon, spectral slopes are aligned with theoretical expectations for buoyancy-driven convection, suggestive of the occurrence of a thick layer of well-developed convective turbulence near the seafloor.

Constructing a phase-averaged view of a semidiurnal tidal cycle. To draw statistically robust relationships between mixing and its tidal drivers, and elucidate implications for upwelling, we construct a phase-averaged picture of the temporal evolution of the along-canyon flow (Fig. 3a), χ (Fig. 3b) and a range of ancillary

parameters (Fig. S5) over a semidiurnal tidal cycle (see Methods). The time series of phase used to generate the averages is calculated by fitting a sinusoid to the up-canyon velocity over a sliding 3-day window. This sinusoid has four free parameters (amplitude, frequency, phase and offset) that are separately fitted via a least squares approach for each window. The resulting fitted phase is then taken as the phase at the mid-point of the window. This approach was employed to capture possible subtle changes in the tidal drivers over the time series. An analogous analysis using a fixed M2 period produces very similar phase-averaged diagnostics, although slightly blurred due to smoothing by small phase errors.

Calculation of diapycnal upwelling from mooring-based temperature observations. The rate of diapycnal (closely equivalent to diathermal within the canyon, see Methods) upwelling, w_T^* , across the canyon's mean stratification is computed from the mooring-based estimates of χ . An expression for w_T^* may be derived by considering the diathermal flux²⁷,

$$\phi = -\kappa \frac{\partial z^*}{\partial T} \langle |\nabla T|^2 \rangle_{z^*},$$

where z^* is the distance in the diathermal direction and $\langle \rangle_{z^*}$ denotes an average over a temperature surface in a bounded volume. Combining this equation with the definition of $\chi = 2\kappa \langle |\nabla T|^2 \rangle$ (where the angled brackets indicate an average over a local volume, with spatial scales comparable to those of three-dimensional turbulence; ref. 42) and expressing w_T^* as the rate of change of the diathermal flux in the diathermal direction yields:

$$w_T^* = -\frac{\partial \phi}{\partial z^*} = \frac{1}{2} \frac{\partial (\langle \chi \rangle_{z^*} \frac{\partial z^*}{\partial T})}{\partial z^*} = \frac{1}{2} \frac{\partial \langle \chi \rangle_{z^*}}{\partial T} + \frac{1}{2} \langle \chi \rangle_{z^*} \frac{\partial^2 z^* / \partial T^2}{\partial z^* / \partial T}.$$

w_T^* is given in units of $^{\circ}\text{C s}^{-1}$. Its definition encapsulates the notion that a water parcel mixing with surrounding waters of diverse temperatures will translate toward the temperature class with which it mixes most intensely^{5,6}. Note that, since isopycnal gradients of temperature are negligible within the canyon, the in-canyon temperature variance is overwhelmingly produced by diapycnal – and not isopycnal – flows, such that w_T^* is an accurate metric of the diapycnal velocity. w_T^* can be expressed in the more familiar units of m s^{-1} via scaling with the diathermal (close to vertical within the canyon) gradient of potential temperature, i.e. $w^* \approx w_T^* / \frac{\partial T}{\partial z^*}$.

Our calculation of w_T^* from mooring-based estimates of χ makes two inevitable assumptions. First, z^* is approximated by the local vertical direction, i.e. the only direction sampled by the mooring. This approximation is supported by a comparison of directly-measured vertical temperature gradients with indirectly-estimated along-canyon temperature gradients (inferred via application of Taylor’s frozen field hypothesis, by which $\frac{\partial T}{\partial x} \sim \frac{\partial T}{\partial t} / U$ with U as the ADCP-measured speed of the along-canyon flow past the temperature sensor). This comparison suggests that the characteristic magnitude of vertical gradients exceeds that of horizontal gradients by one-to-three orders of magnitude on a wide range of time scales longer than those of the inertial subrange (i.e. $\gg 1$ min), such that the diathermal direction is close to vertical on those time scales.

Second, $\langle \chi \rangle_{z^*}$ (i.e. the isothermal average of χ within the bounded volume of our calculation at any specific tidal phase) is assumed to be adequately represented by χ as sampled by the mooring at that phase. There is no *a priori* justification for this approximation; rather, its adoption places subtle constraints on the interpretation of our

w_T^* diagnostics. The ‘bounded volume’ to which these diagnostics apply may be regarded as a body of water centred at the mooring site, extending up to ~6 km (double the along-canyon semidiurnal tidal excursion) along the canyon’s thalweg, and sloshing up- and down-canyon by up to ~3 km. Since the semidiurnal tidal cycle of χ exhibits the same characteristic patterns at different locations along the canyon (e.g., compare Figures 2c and S3d), our estimated w_T^* at any tidal phase could be thought of as the upwelling rate experienced by the volume at that phase. This would be the case even if different parts of the volume are subject to different tidal phases at any instant in time, such as might occur if the baroclinic tide’s propagation along the canyon takes a substantial fraction of a semidiurnal period.

The calculation of w_T^* is performed in three steps. First, a vertical profile of χ is constructed at intervals of 5 minutes by bin-averaging all the χ estimates from individual thermistors within a half-bin period of each time bin centre. Second, each vertical profile of χ is re-mapped onto temperature space by bin-averaging in temperature cells at intervals of 0.1°C. Third, $\partial\chi/\partial T$, $\partial z^*/\partial T$ and $\partial^2 z^*/\partial T^2$ are computed in each time bin and temperature class via centred differences, with z^* approximated by the vertical coordinate. This step assumes that a single vertical mooring captures the bulk of the temperature dependence of χ . The resulting estimate of w_T^* is primarily determined by the $\frac{1}{2} \frac{\partial\langle\chi\rangle_{z^*}}{\partial T}$ term, although the contribution from the $\frac{1}{2} \langle\chi\rangle_{z^*} \frac{\partial^2 z^*/\partial T^2}{\partial z^*/\partial T}$ term is also significant and reinforces the patterns in the dominant term (Fig. S6).

To gauge the uncertainty in w_T^* related to temporal variability and instrumental noise, we compute the interquartile range of w_T^* estimates obtained from many phased-

averaged χ fields, each constructed from a set of three semidiurnal tidal periods (the smallest record length required to map out χ as a function of potential temperature and tidal phase) (Fig. 3d). This suggests that, at a minimum, the near-boundary upwelling signal of w_T^* in waters colder than $\sim 5^\circ\text{C}$ is robust and persistent during the period of mooring observations. Although errors in the thermistor-based estimates of χ (discussed above) may potentially introduce further quantitative uncertainty in w_T^* , such errors are too modest to affect our conclusions on near-boundary upwelling.

Origin of the along-canyon shear. The along-canyon shear exerts an important influence on the intensity of upwelling within the canyon, as it is this shear that triggers convective overturns by differentially advecting denser waters over lighter waters during flood tide and resetting a stable stratification during the ebb phase (Fig. S1). To gain some insight into the along-canyon shear's physical origin, Figure S4a shows the power density spectrum of the ADCP-measured along-canyon velocity shear, estimated from a linear fit to the full vertical profile of the along-canyon velocity. The spectrum exhibits peaks at the K1 and M2 tidal frequencies and their harmonics, with little energy at the higher frequencies related to small-scale turbulence. Further, the along-canyon shear is much more energetic than the cross-canyon shear (not shown). This confirms that the shear is associated with along-canyon baroclinic tidal motions within the canyon.

Next, we wish to assess whether the tidal shear results from turbulent motions arresting the near-bottom flow – as might be expected from the action of topographic drag – or from the propagation of waves (e.g., baroclinic tides) within the canyon. To illuminate this issue, Figure S4b displays the coherence between the along-canyon and vertical velocities, which provides a metric of the effectiveness of motions of a given frequency

in the vertical redistribution of along-canyon momentum. Such redistribution is arguably required to explain the emergence of the tidal shear. Coherence is found to be large exclusively at tidal frequencies, and not at the higher frequencies of small-scale turbulent flows. This strongly suggests that the tidal shear is associated with baroclinic waves that arise in the canyon^{28,29}, and not the result of a turbulent-frequency momentum flux divergence. The observed dominance of the shear's along-canyon component further suggests that the waves' dispersion relationship is Kelvin-like⁶⁰, in that cross-canyon motions are suppressed by the presence of supercritical lateral walls. Future work will address the question of how these waves are generated.

Although the ADCP measurements used in this analysis do not resolve the full spectrum of small-scale turbulent motions, the above results are corroborated through an analogous examination of MAVS current meter observations from the same mooring (not shown), which have much finer temporal resolution and comprehensively capture the inertial subrange.

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787

788 **Author Contributions** A.C.N.G. and C.P.S. performed analyses, produced figures and
789 wrote the manuscript, with feedback and suggestions from all authors. A.C.N.G., G.V.,
790 H.M., K.L.P., M.H.A., M.J.M. and R.F. conceived and proposed the BLT-Recipes
791 experiment. A.C.N.G., A.F., B.F.C., B.L.W.C., C.P.S., G.V., H.M., H.v.H., K.L.P.,
792 M.H.A., M.J.M. and N.C. contributed to the design and execution of the experiment.
793 Y.M. and R.F. led the exploration of the origin of the in-canyon shear.

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795 **Competing Interests** The authors declare no competing interests.

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798 www.nature.com/reprints. Correspondence and requests for materials should be
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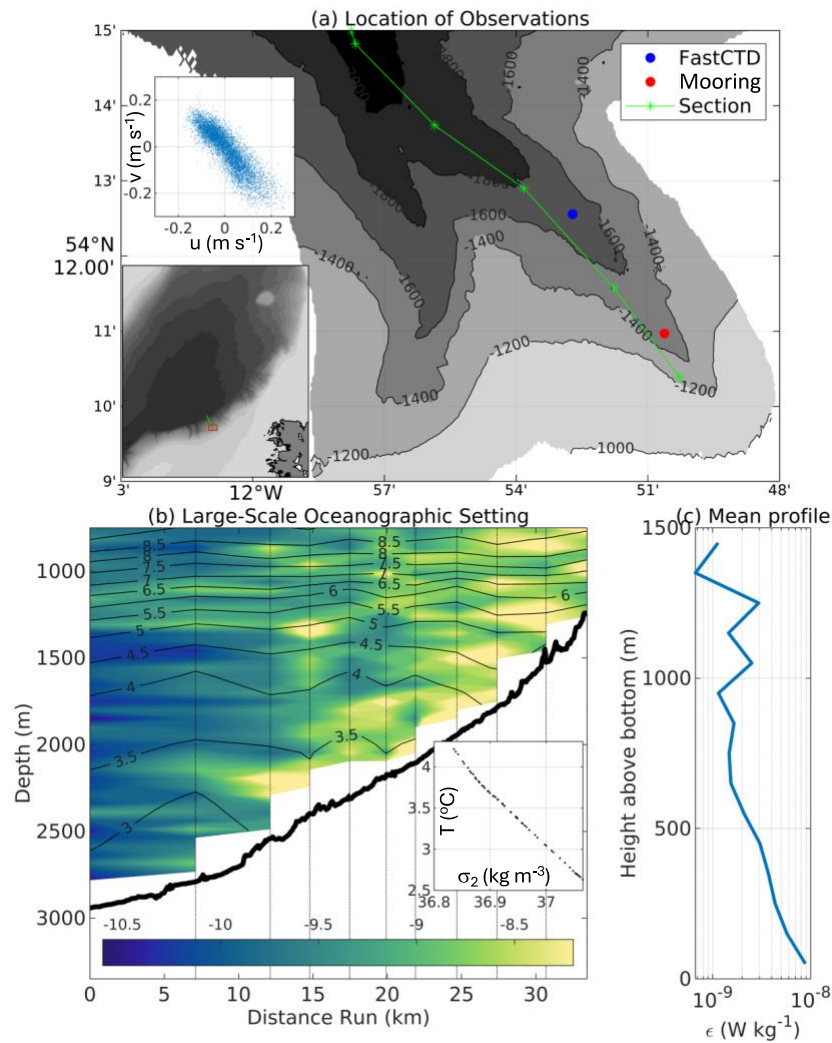


Figure 1 | Overview of BLT-Recipes experimental domain and observations. (a) Topography of the Rockall Trough in the Northeast Atlantic (lower inset), and of the continental-slope canyon measured during the BLT-Recipes experiment (main panel) with measurement locations indicated by instrument-referencing symbols. The near-bottom (averaged over a height above bottom range of 50-300 m) velocity at the location of the mooring is shown in the upper inset, and indicates that tidal fluctuations in the in-canyon flow are predominantly oriented along the canyon. (b) Rate of turbulent kinetic energy dissipation (ϵ , shading) and potential temperature (contours) along the canyon's thalweg (section marked in Fig. 1a). The inset illustrates the potential temperature (T) - density relationship observed within the canyon, where the

812 density variable shown (σ_2) is the potential density relative to 2000 dbar. (c) Along-
813 canyon-mean profile of ε versus height above bottom.

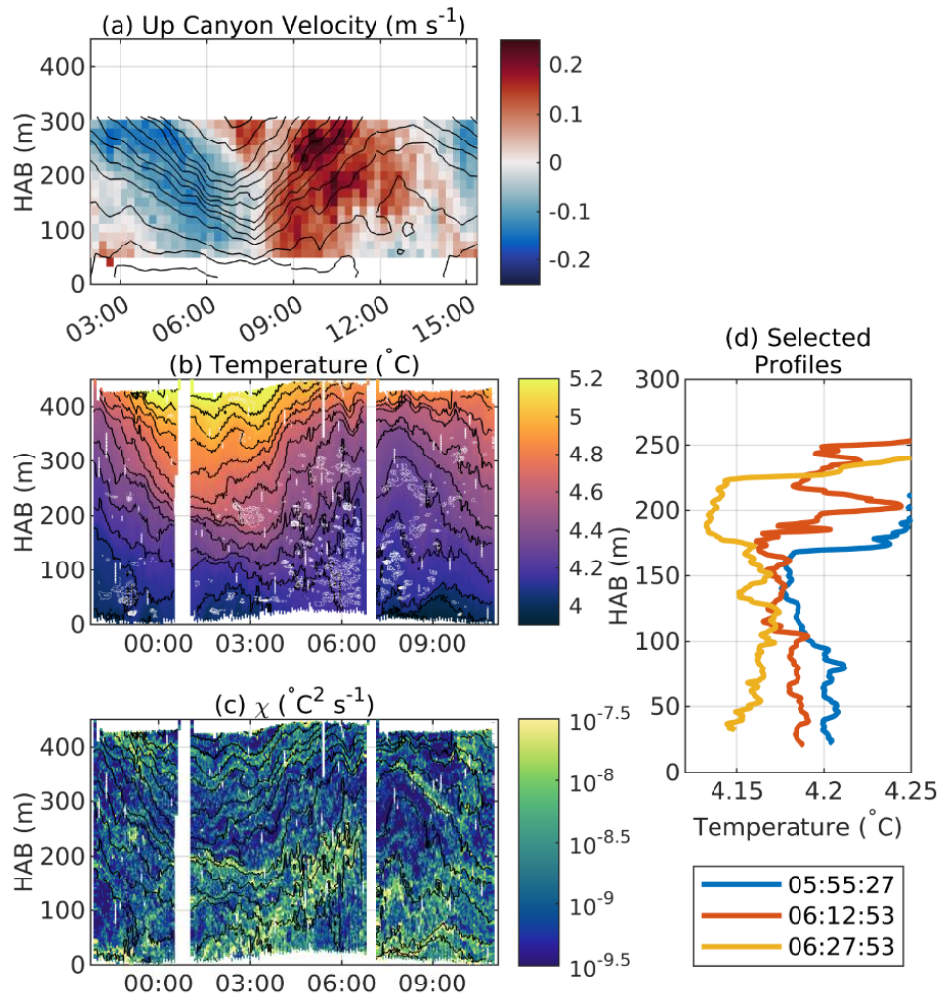
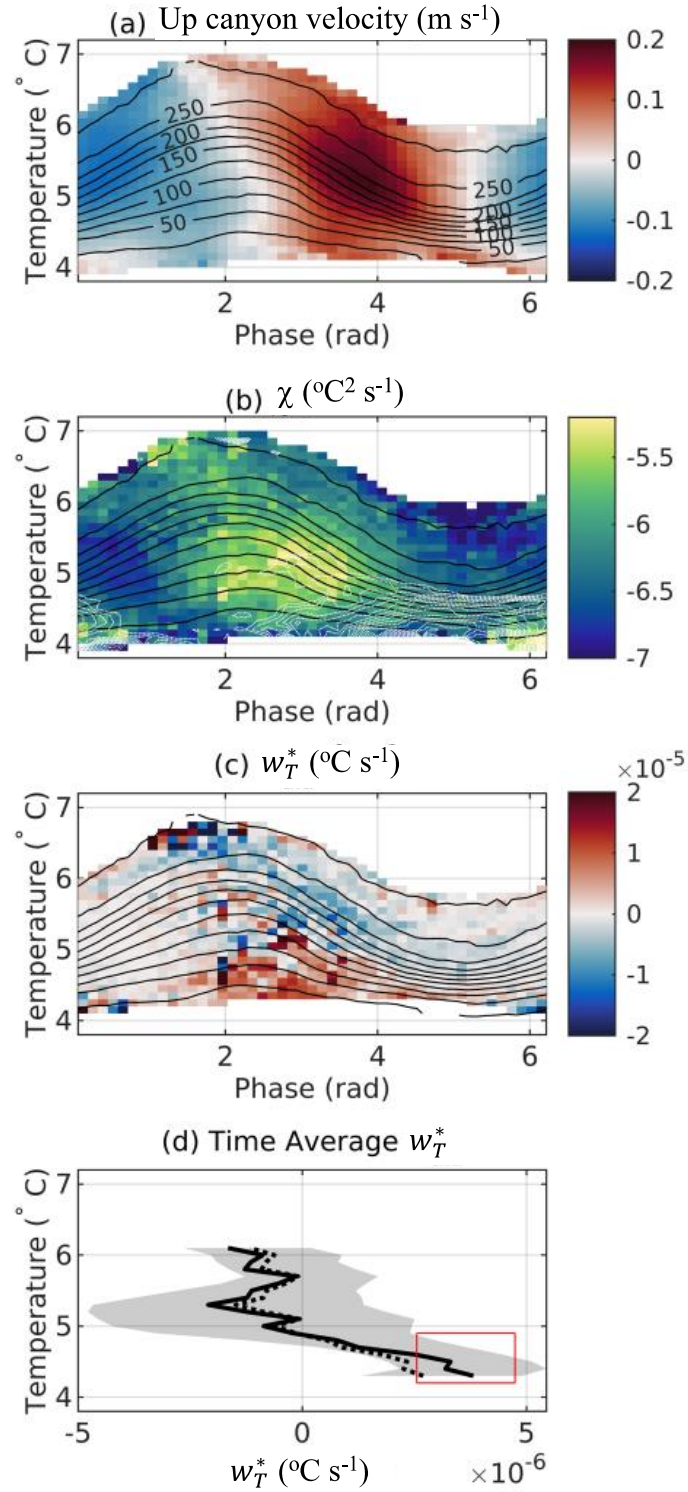


Figure 2 | A semidiurnal tidal cycle of along-canyon flow, hydrographic structure and turbulent mixing. (a) Along-canyon velocity (shading, positive values indicate up-canyon flow) and potential temperature (contours) measured by the mooring's ADCP and thermistors, respectively, on 8 July 2021. HAB: height above bottom. (b) Potential temperature (shading and black contours) measured by the FastCTD on 2-3 July 2021. White contours indicate occurrences of a statically unstable vertical temperature gradient (i.e. $\partial T/\partial z < 0$). (c) Rate of turbulent dissipation of temperature variance (χ , shading) measured by the FastCTD, with potential temperature contours overlaid in black. Panels (a) and (b)-(c), which show data from different periods at approximately the same location, have been aligned by the phase of the semidiurnal tide. (d) Selected profiles of potential temperature during the period with extensive statically unstable conditions.



827

828 **Figure 3 | Phase-averaged semidiurnal tidal cycle of along-canyon flow, turbulent**
 829 **mixing and diapycnal upwelling.** Phase-averaged (a) up-canyon velocity, (b) rate of
 830 turbulent dissipation of temperature variance (χ), and (c) diapycnal velocity (w_T^*) –
 831 indicated by shading in each panel – as a function of potential temperature for a

832 characteristic semidiurnal tidal cycle (see Methods). Height above bottom contours are
833 overlaid in black and labelled in (a). (d) Mean (across all phases) profile of diapycnal
834 velocity as a function of potential temperature. The solid and dashed lines respectively
835 show the diapycnal velocity calculated using dissipation ratio (Γ) values of 0.2 and 0.4
836 (see Methods). The grey shading provides a measure of the uncertainty in w_T^* related to
837 temporal variability and instrumental noise. This is computed as the interquartile range
838 of w_T^* estimates obtained from many phased-averaged χ fields, each constructed from
839 a set of three semidiurnal tidal periods. The diapycnal velocity determined from the
840 dye¹² is added as a rectangle (with horizontal and vertical side lengths respectively
841 indicating the uncertainty and the potential temperature range sampled by the dye,
842 where this range has been shifted toward warmer waters by the difference between the
843 coldest temperatures sampled by the dye and the mooring) for comparison.

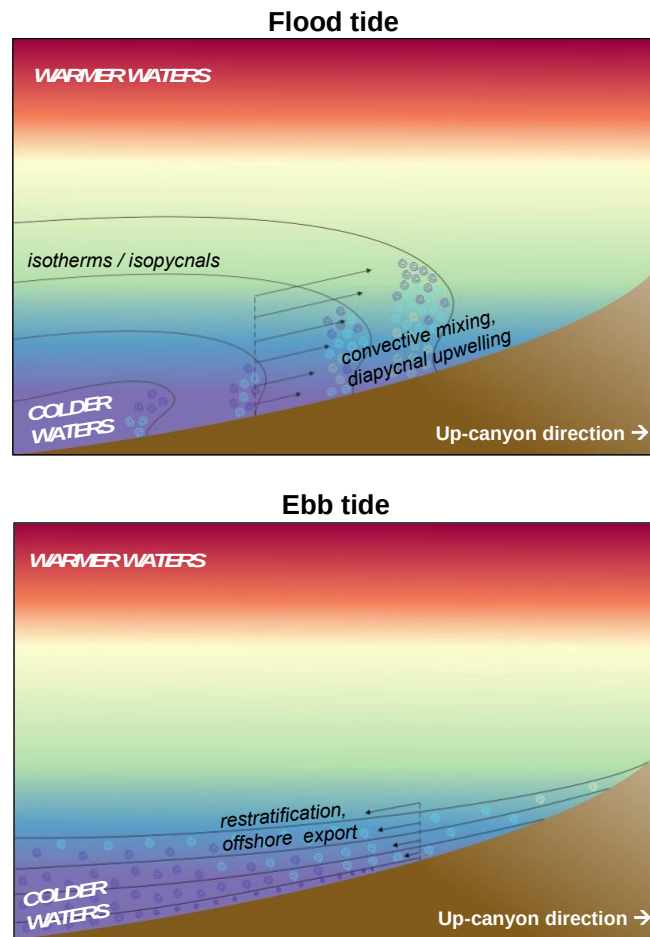
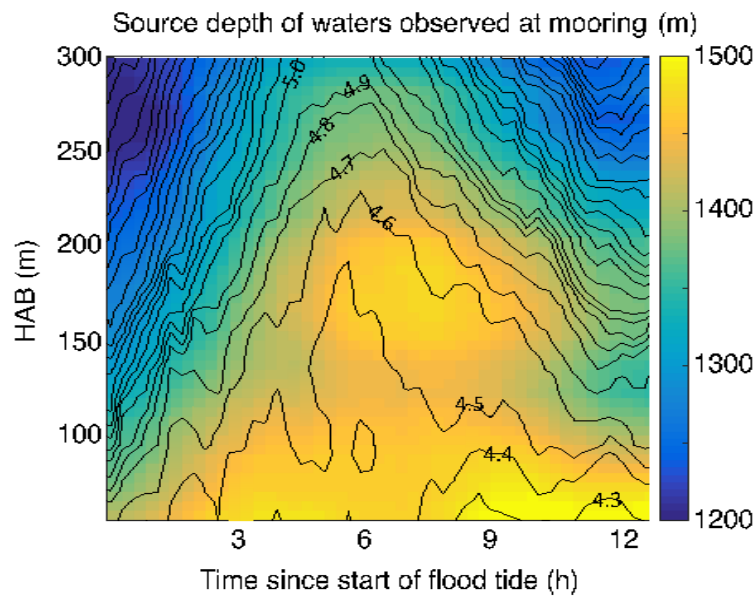
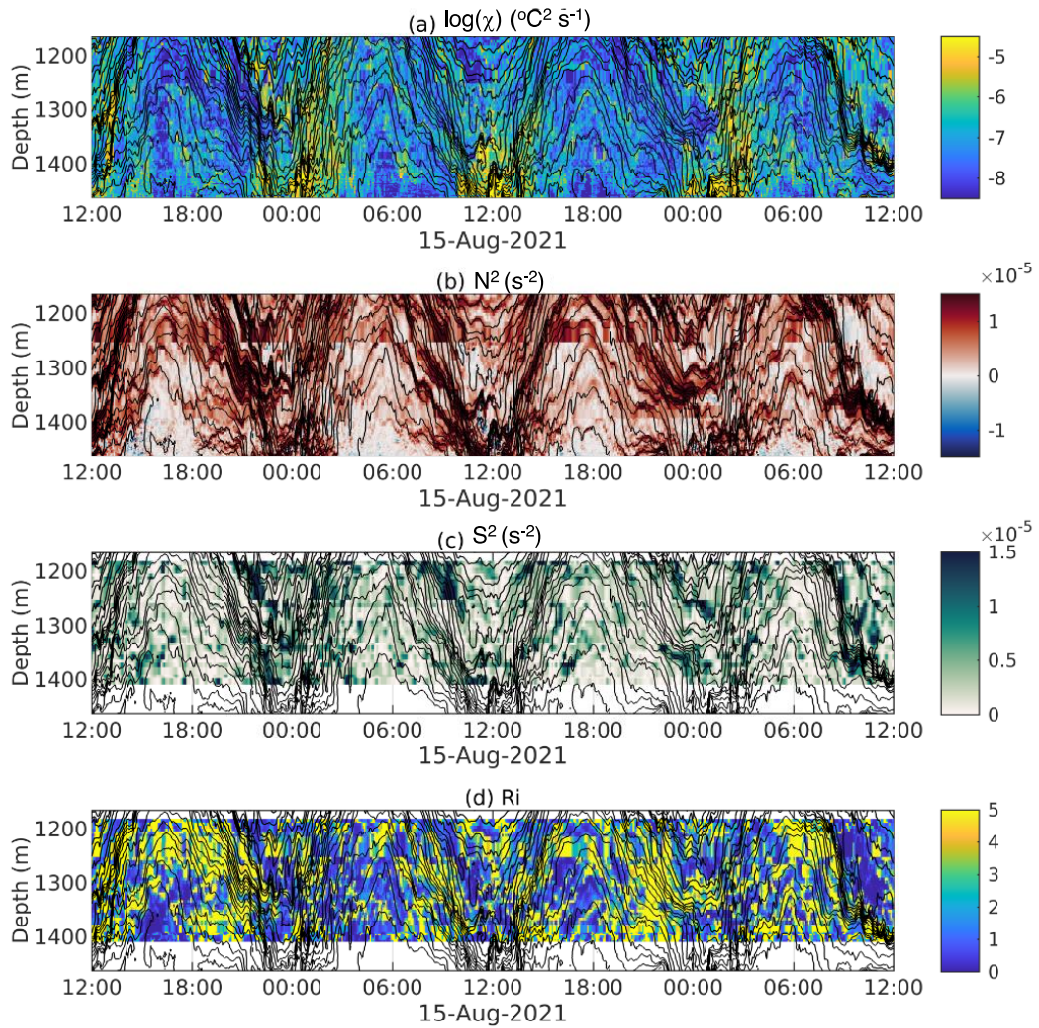


Figure 4 | Schematic of along-boundary upwelling driven by tidally-generated convective mixing. (a) During flood tide, the sheared up-canyon flow causes relatively dense waters aloft to overshoot and convectively mix with lighter near-bottom waters, leading to diapycnal upwelling of the dense waters along the boundary. (b) During ebb tide, the intensification of down-canyon flow with height induces re-stratification within the canyon, and supports the offshore export of waters that were newly mixed and upwelled during the previous flood tide.



852

853 **Figure S1 | Analytical model of the role of vertically differential along-canyon**
 854 **advection in generating convective conditions.** Source depth (shading) of waters
 855 sampled by the mooring, estimated with an idealized analytical model of the advection
 856 of the in-canyon vertical temperature profile at the start of flood tide by the measured
 857 along-canyon flow. Comparison with the observed potential temperature record
 858 (contours) indicates that the measured emergence of a thick, homogeneous BBL in the
 859 deepest ~250 m during flood tide stems from all BBL waters having originated at
 860 approximately the same depth upstream, thus priming the layer for convection.
 861 Conversely, BBL thinning coincides with a progressive sharpening of the vertical
 862 gradient in the source depth of near-bottom waters. HAB: height above bottom.



863

864 **Figure S2 | A 2-day example of semidiurnal cycles of turbulent mixing, in-canyon**

865 **stratification, shear and Richardson number.** (a) Rate of turbulent dissipation of

866 temperature variance (χ , shading) and potential temperature (contours) during a 2-day

867 subset of the mooring record. (b) Squared buoyancy frequency (N^2 , shading) and

868 potential temperature (contours) during the same period. (c) Squared shear (S^2 , shading)

869 and potential temperature (contours). (d) Richardson number (Ri , shading) and

870 potential temperature (contours).

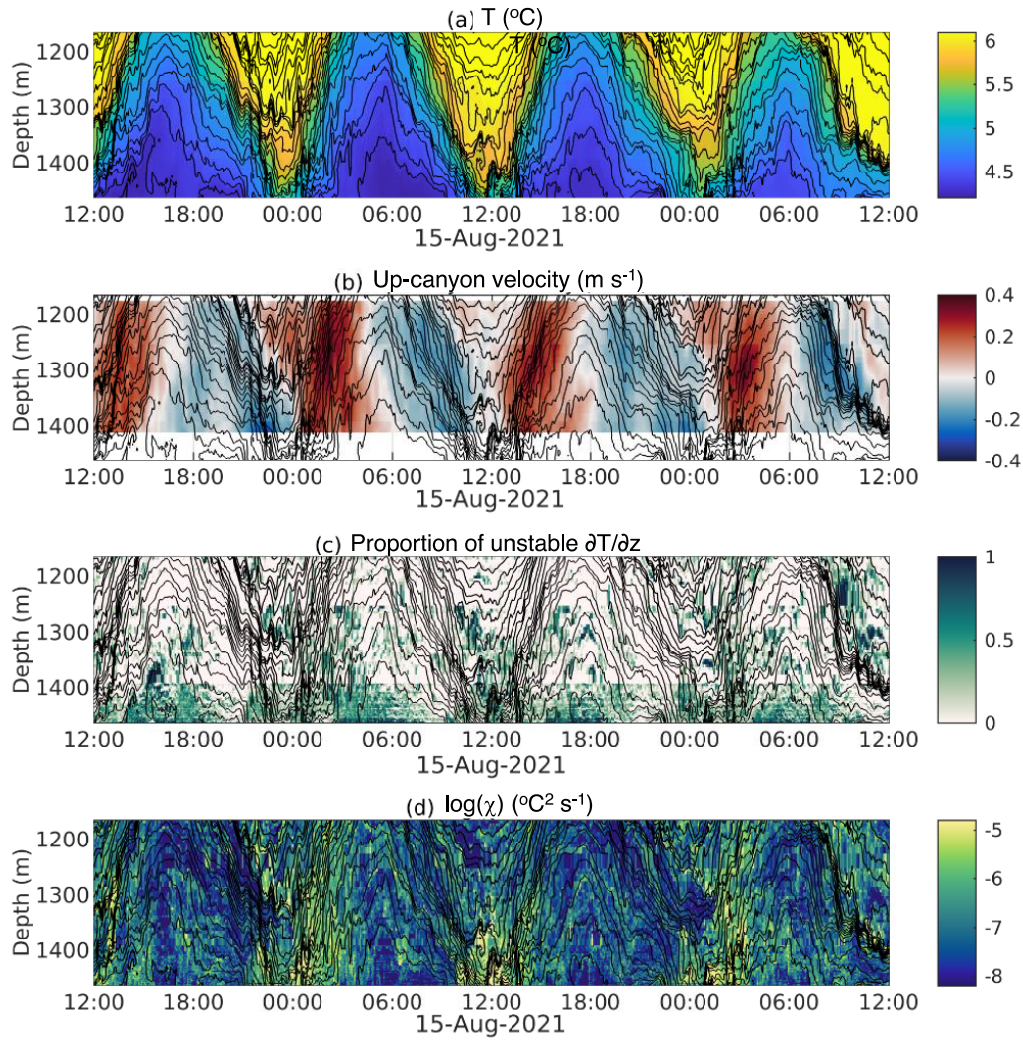


Figure S3 | A 2-day example of semidiurnal cycles of hydrographic structure, along-canyon flow and turbulent mixing. (a) Potential temperature (shading and black contours) during a 2-day subset of the mooring record. (b) Along-canyon velocity (shading) and potential temperature (contours) in the same period. (c) Fractional occurrence of a statically unstable vertical temperature gradient (i.e. $\partial T/\partial z < 0$) in the 1 Hz temperature data within each 15-min bin (shading), with potential temperature contours overlaid in black. A fraction of 1 indicates sustained unstable conditions within a bin, and a fraction of 0 denotes continuous stability during that time. (d) Rate of turbulent dissipation of temperature variance (χ , shading) and potential temperature (contours).

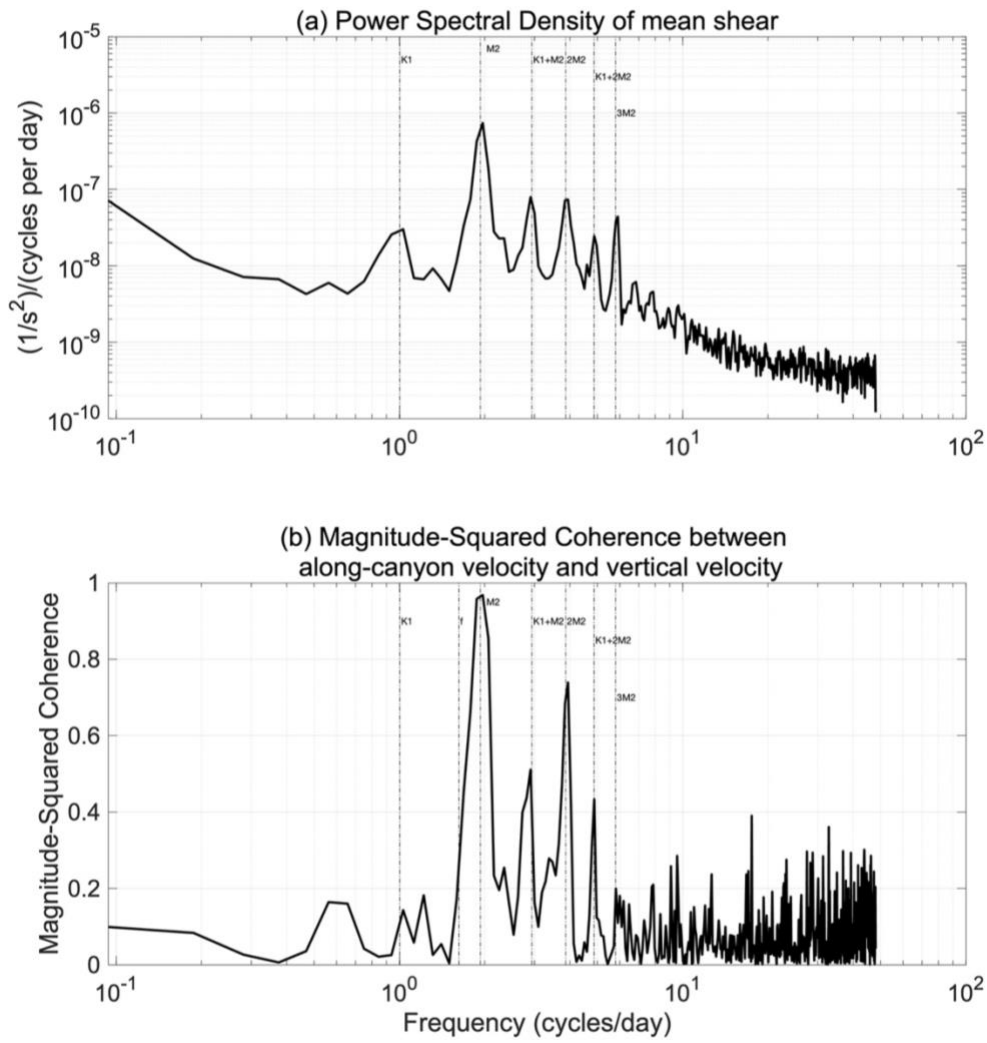


Figure S4 | Investigation of the physical origin of along-canyon shear. (a) Power spectral density of the (ADCP-measured) mean along-canyon velocity shear, estimated from a linear fit to the full vertical profile of the along-canyon velocity. (b) Magnitude-squared coherence between the along-canyon velocity and the vertical velocity. The K1 and M2 tidal constituents and their harmonics are indicated by dashed lines and labelled in both panels.

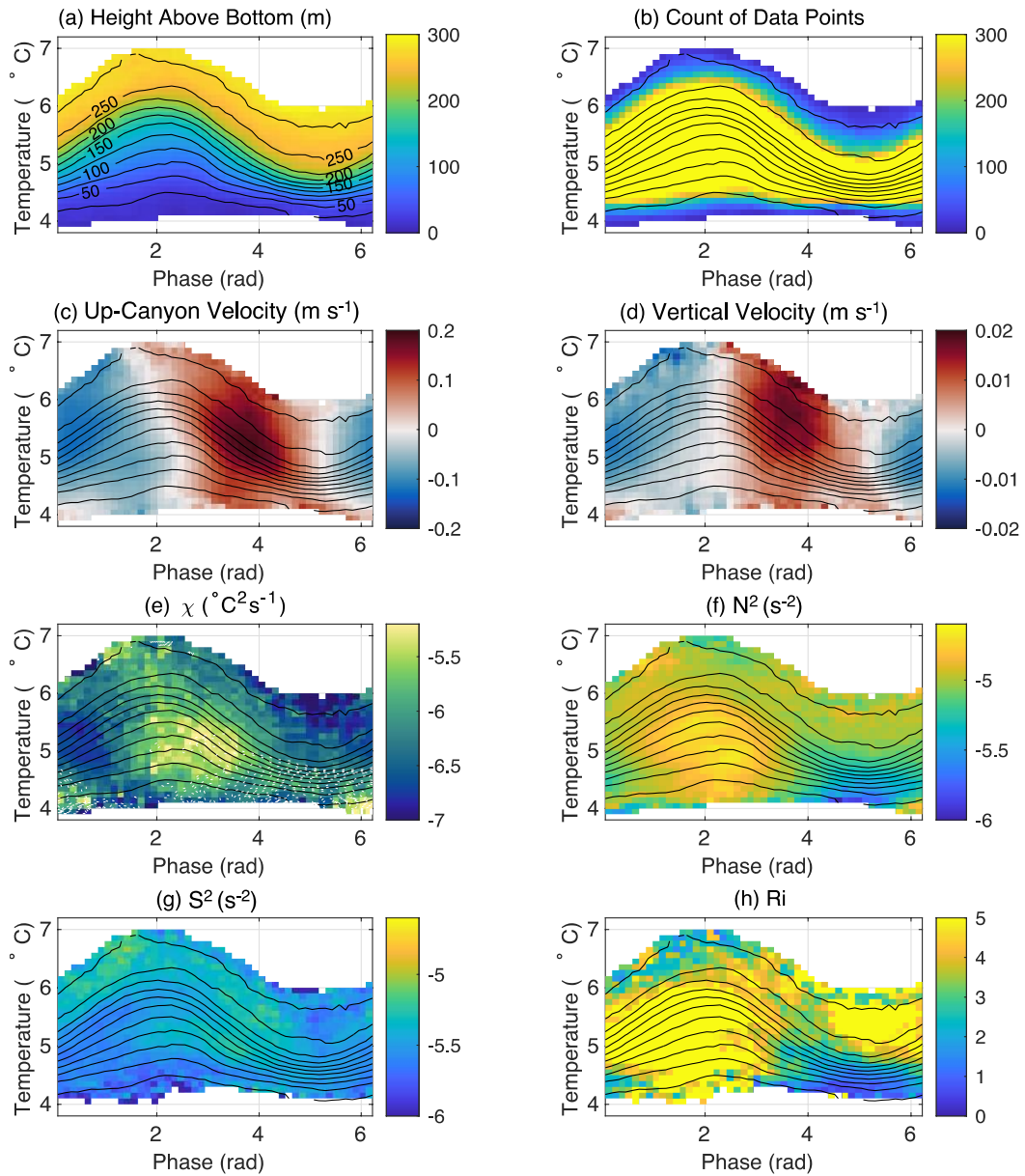
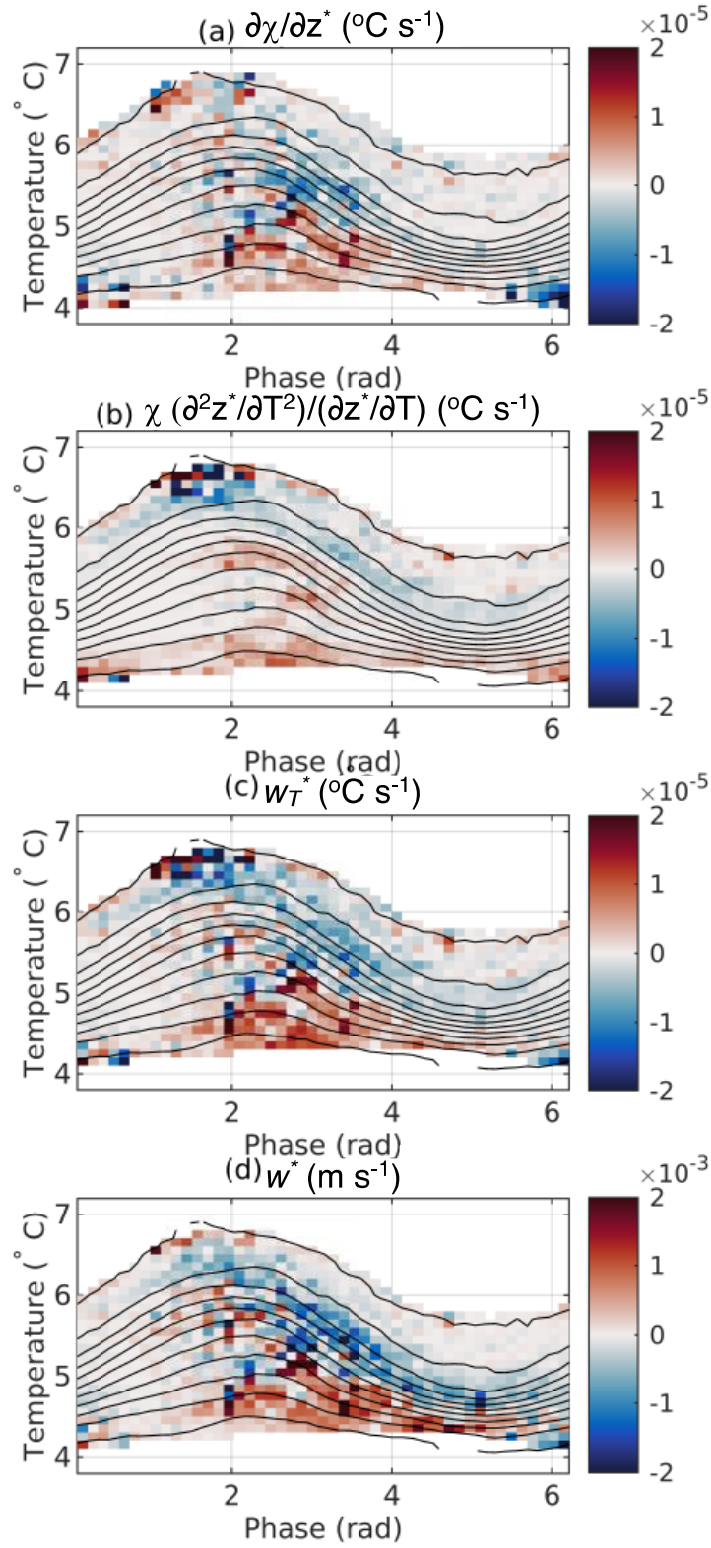


Figure S5 | Phase-averaged semidiurnal tidal cycle of in-canyon flow, hydrographic structure and turbulence parameters. Phase-averaged (a) height above bottom of isotherms; (b) data density; (c) up-canyon velocity; (d) vertical velocity; (e) rate of dissipation of temperature variance (χ); (f) squared buoyancy frequency (N^2); (g) squared shear (S^2); and (h) Richardson number (Ri) – as a function of potential temperature for a characteristic semidiurnal tidal cycle (shading; see Methods). Height above bottom contours are overlaid in black and labeled.



898

899 **Figure S6 | Phase-averaged semidiurnal tidal cycle of diapycnal upwelling and its**

900 **constituent terms.** Phase-averaged (a) $\frac{\partial \chi}{\partial T}$ term of diapycnal velocity, (b) $\chi \frac{\partial^2 z^* / \partial T^2}{\partial z^* / \partial T}$

901 term of diapycnal velocity, (c) diapycnal velocity expressed as a warming rate (w_T^*),

902 and (d) diapycnal velocity in m s^{-1} ($w^* \approx w_T^* / \frac{\partial T}{\partial z}$) – indicated by shading in each panel

903 – as a function of potential temperature for a characteristic semidiurnal tidal cycle (see
904 Methods). Height above bottom contours are overlaid in black.

Supplementary Movie | An 89-day mooring record of along-canyon flow, hydrographic structure and turbulent mixing. (a) Up-canyon velocity (shading) and potential temperature (contours) during the 89-day mooring record. (b) Potential temperature (shading and contours) in the same period. (c) Rate of turbulent dissipation of temperature variance (χ , shading), with potential temperature contours overlaid in black.

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